Twitter vs X: A Comparative Analysis of Moderation, Sentiment, and Freedom of Speech in the Era of Platform Transition

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***Abstract*—Since Elon Musk took charge of Twitter by transforming it into X users started speculating about how content moderation policies would change alongside freedom of speech and platform sentiment. This study presents an empirical investigation of this transition by comparing over 300,000 tweets from both the pre-acquisition "Twitter" and total tweets in the dataset of post-acquisition "X" periods. Using advanced natural language processing (NLP) techniques, including Roberta-based sentiment classification, VADER lexicon analysis, and BERTopic topic modeling, we evaluated variations in sentiment, moderation discourse, and thematic diversity. The researchers utilized an 80+ keyword list specifically related to moderation to extract appropriate tweets which allowed them to compare sentiment across different social media platforms. Research data indicates a substantial change in X users' content toward debating ideology and technology but does not show significant negative sentiment rises despite its different content monitoring rules. Major corporate governance transformations create subtle shifts in digital media discourse which this study contributes to recording within academic research.**

***Index Terms*—Index Terms—Twitter, X, sentiment analysis, content moderation, BERTopic, RoBERTa, social media discourse, NLP**

I. Introduction and Statement of the Problem

Elon Musk's acquisition of Twitter and subsequent rebranding to "X" has generated widespread controversy surrounding the site's degree of dedication to content moderation, user agency, and the broader implications for public discourse. Elon Musk's position as a "free speech absolutist" has instigated sweeping changes in leadership, including policy reforms, reduction of moderation staff, and the removal of bans on previously suspended accounts. Critics and admirers were uncertain whether these structural reforms would result in more disinformation, hate speech, or less moderation enforcement.

Identify applicable funding agency here. If none, delete this.

This project investigates these issues through a comprehensive natural language processing (NLP)-based study of user discourse before and after Twitter's transformation. We analyze not only surface-level sentiment but also the deeper thematic structures of tweets referencing moderation, censorship, and platform policies. By comparing moderation-related tweets between Twitter and X, we aim to answer whether sentiment polarity has shifted, whether topical diversity has increased, and whether content moderation discourse has changed in tone or frequency.Twitter vs X: A Comparative Analysis of Moderation, Sentiment, and Freedom of Speech in the Era of Platform Transition

The central research problem is thus: *Has the platform-wide sentiment and topic discourse surrounding content moderation significantly changed between Twitter and X?* To address this, we use machine learning methods to analyze lexical structures, sentiment distributions, and topic models from curated tweet datasets representing each era of the platform.

II. Review of Literature

Several studies have addressed the impact of social media platform governance on content moderation, user sentiment, and the dynamics of public discourse. Existing literature offers frameworks to understand platform policy changes and their consequences for online behavior.

Ferrara (2020) explored the rise of social bots and misinformation on Twitter, noting that governance models significantly influence content credibility and discourse quality [5]. In parallel, Antypas et al. (2022) introduced transformer-based topic modeling approaches suitable for capturing nuanced themes on short-form text platforms like Twitter, establishing BERTopic as an effective tool for this type of analysis [1].

Wang and Liu (2023) demonstrated the benefits of combining sentiment analysis with topic modeling to explore public attitudes during sociopolitical events, showing how sentiment polarity can shift based on governance, news cycles, and platform features [3]. Roychowdhury (2023) further emphasized that sentiment analysis tools like Roberta outperform lexicon-based models such as VADER for complex linguistic structures and sarcasm, often found in tweets [4].

Qayyum et al. (2024) specifically examined toxic discourse patterns on Twitter and observed how platform policy changes correlated with shifts in negative sentiment, particularly among previously banned users or ideological groups [2]. Their work provides evidence that moderation policies are directly tied to sentiment shifts and topic realignments.

The current study builds upon these findings by combining sentiment analysis, keyword filtering, and BERTopic modeling to analyze platform-wide transitions. Unlike earlier research that focuses on isolated events, this study takes a broader comparative approach across two temporal versions of the same platform—Twitter and X—focusing exclusively on moderation-related discourse.

III. Objectives of the Study

The primary aim of this study is to assess whether the shift from Twitter to X resulted in measurable changes in sentiment and thematic discourse, particularly regarding content moderation. The study is designed to explore multiple dimensions of user expression and platform response post-transition.

The specific objectives are:

- To measure and compare sentiment distributions across the pre-acquisition (Twitter) and post-acquisition (X) platforms using both lexicon-based and transformer-based approaches.
- To detect moderation-related tweets using a manually curated keyword set, enabling focused analysis on governance-related discourse.
- To apply topic modeling via BERTopic on moderation-specific tweets and compare dominant themes between Twitter and X.
- To visualize content patterns and lexical shifts using tweet-length distributions and word clouds.
- To test the statistical significance of sentiment variation using the Chi-square test across moderation discourse.

These objectives support a comprehensive understanding of how platform governance impacts user expression, discourse tone, and thematic alignment.

IV. System Architecture

The architecture diagram shown in Figure 1 outlines the overall pipeline of the project, starting from the acquisition and cleaning of datasets, followed by preprocessing, sentiment classification using VADER and RoBERTa, keyword-based filtering for moderation-related tweets, and finally BERTopic-based topic modeling and analysis. This structured flow ensures both quantitative and thematic comparison of moderation discourse across Twitter and X.

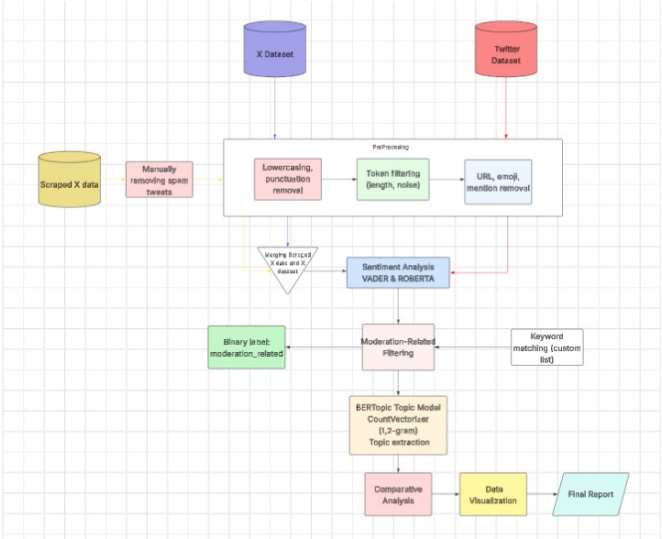

Fig. 1. Workflow Architecture: From Data Collection to Comparative Analysis

V. Data Collection

Two distinct datasets were collected to represent the pre- and post-acquisition phases:

- **Twitter Dataset (Pre-Elon Musk):** A collection of 300,000 tweets posted prior to Elon Musk's takeover, sourced from Kaggle archives and historical tweet scrapers.
- **X Dataset (Post-Elon Musk):** A similarly sized dataset of tweets posted after the rebranding to X, collected using public APIs and Hugging Face datasets.

Both datasets underwent cleaning procedures including:

- Removal of retweets and duplicates.
- Lowercasing and punctuation stripping.
- Tokenization and stopword removal.

Moderation-related tweets were identified using a curated keyword list consisting of 80+ terms, categorized into six thematic buckets:

- Moderation & Restrictions (e.g., "ban", "shadowban")
- Free Speech & Censorship (e.g., "freedom of speech", "cancel culture")
- Policy References (e.g., "terms of service", "violated guidelines")
- Disinformation & Propaganda (e.g., "misinformation", "suppression")
- Elon/Ownership-Specific (e.g., "elon musk", "musk takeover")
- NSFW/Extremes (e.g., "abuse", "child porn", "graphic content")

This list was used to flag and filter tweets relevant to content moderation, ensuring both platforms were analyzed consistently.

```
moderation_keywords = [
    # Moderation & Restrictions
    "ban", "banned", "suspend", "suspended", "shadowban", "shadow banned", "account locked",
    "restricted", "rate limited", "deplatform", "muted", "post removed", "account flagged",
    "removed", "deleted", "violation", "restrict",
    "account locked", "blocked",

    # Free Speech & Censorship Discussion
    "freedom of speech", "free speech", "censorship", "silenced", "cancel culture",
    "narrative control", "content moderation", "fact check", "flagged tweet", "moderation bias",

    # Policy and Guidelines References
    "community guidelines", "terms of service", "policy violation", "violated guidelines",
    "platform rules", "automated moderation", "ai moderation", "trust and safety",

    # Suppression, Propaganda, Disinformation
    "suppression", "disinformation", "misinformation", "propaganda", "biased moderation",

    # Elon/Ownership-specific
    "elon musk", "elon changed twitter", "under elon", "new twitter policy",
    "twitter 2.0", "free speech absolutist", "elon said", "musk takeover",

    # Abuse, NSFW, Extremes
    "hate speech", "abusive", "harassment", "nfsw", "graphic content", "revenge porn",
    "nudity", "porn", "child safety", "self harm", "suicide", "terrorist content", "violent",
    "abuse","harassment","child porn", "child sex", "murder","kill","terrorist", "nude", "sex tape",
    "suicide","i want to die", "massacre", "rape", "isis", "jihad","slut","illegal",
]
```

Fig. 2. Moderation Keywords

VI. Exploratory Data Analysis (EDA) and Hypotheses for the Study

Before applying advanced modeling, we conducted in-depth exploratory analysis on both datasets to understand lexical structures, content diversity, and sentiment polarity.

A. Tweet Length Distribution

The distribution of tweet lengths offers foundational insights into user behavior and platform constraints during each phase. Figure 2 depicts the histogram of tweet lengths in the Twitter dataset, which follows a right-skewed unimodal distribution. The majority of tweets cluster between 10 and 50 characters, peaking around the 25–30 character range. This suggests that tweets during the pre-acquisition period were typically concise, informal, and aligned with Twitter's historical character limit.

In contrast, Figure 3 displays the tweet length distribution in the X dataset. While a similar concentration of tweets exists in the shorter length range (under 100 characters), there is a noticeable long tail extending beyond 500, and in some cases, surpassing 4000 characters. This reflects a fundamental shift in platform dynamics post-acquisition—where longer tweets, likely from premium or verified users with expanded character limits, became more frequent. The presence of such outliers indicates a broader allowance for long-form discourse and possibly an evolution toward more detailed or technical content.

The comparative analysis of these distributions suggests a shift from microblogging to extended expression on X, possibly influenced by changes in policy, verification incentives, and audience targeting strategies.

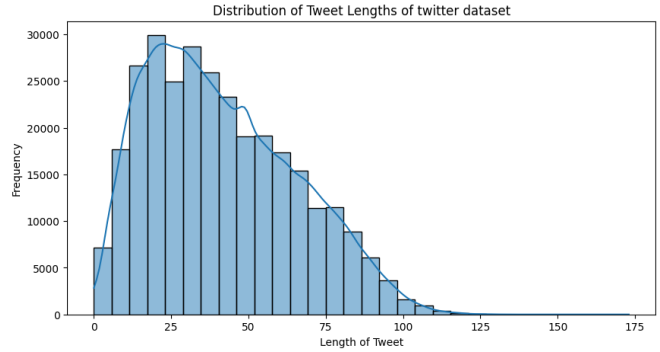

Fig. 3. Distribution of Tweet Lengths – Twitter

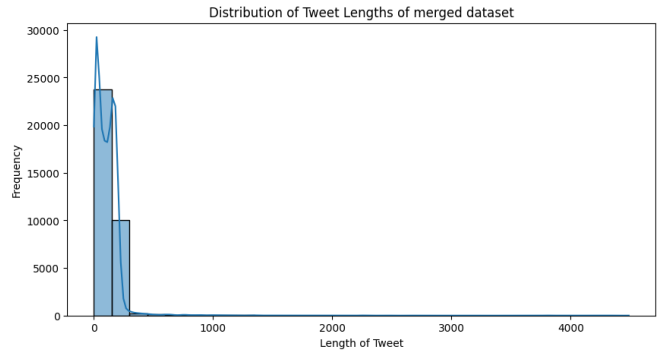

Fig. 4. Distribution of Tweet Lengths – X

B. Word Clouds (Overall Vocabulary)

The word cloud visualizations reveal a distinct lexical contrast between the Twitter and X datasets, showcasing how user discourse evolved following the platform's acquisition and rebranding.

In the Twitter dataset (Figure 4), prominent terms include *"love," "thank," "haha," "work," "gonna," "today," "friend," "time," "wait," "lol,"* and *"amp"*. These words indicate a highly casual, conversational tone, suggesting that the platform at the time served largely as a space for informal expression, social interactions, humor, and emotional outbursts. The frequent use of slang and expressive tokens such as "haha," "lol," and "gonna" underscores Twitter's role as a real-time personal microblogging platform.

Conversely, the X dataset word cloud (Figure 5) demonstrates a shift toward technical, ideological, and blockchain-related terminology. Dominant terms include *"bittensor," "AI," "crypto," "decentralized," "subnet," "validator," "market,"* and *"token."* This reflects a noticeable shift in user base or discourse tone — from emotional and interpersonal content to a more information-rich, niche, and innovation-oriented language style. The prevalence of decentralized technology vocabulary further supports the hypothesis that X has attracted a different segment of users, particularly those engaged in Web3, cryptocurrency, and tech governance.

This comparison illustrates not just a lexical evolution, but a likely transformation in platform function and community culture over time.

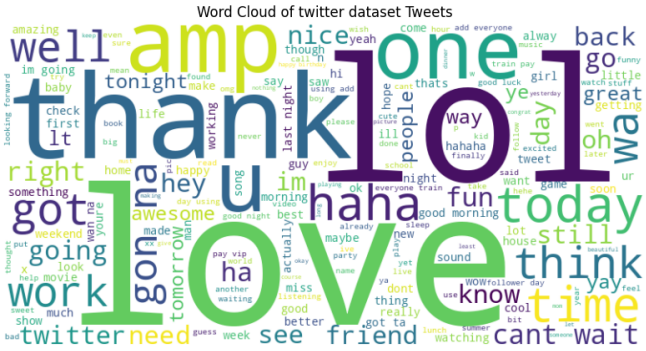

Fig. 5. Word Cloud – Twitter Dataset

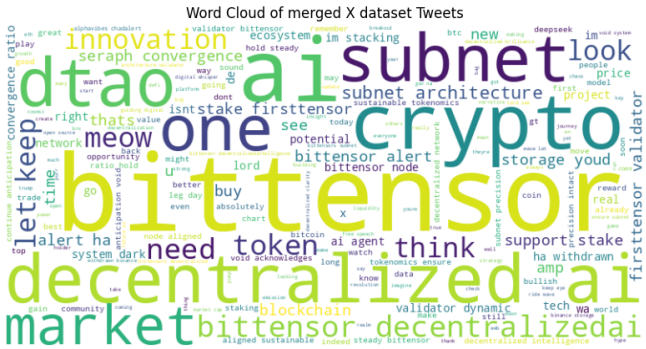

Fig. 6. Word Cloud – X Dataset

C. VADER-Based Sentiment Analysis

To establish a preliminary sentiment baseline, the VADER (Valence Aware Dictionary and sEntiment Reasoner) lexicon-based sentiment analyzer was applied to both the Twitter and X datasets. VADER is optimized for social media text and provides polarity scores that classify texts into three categories: Positive, Neutral, and Negative.

1) Twitter Dataset: As shown in Figure 6, the sentiment distribution for Twitter tweets skews heavily positive:

- **Positive:** Approximately 185,000 tweets
- **Neutral:** Approximately 88,000 tweets
- **Negative:** Approximately 32,000 tweets

This suggests that the Twitter dataset maintained a generally favorable tone, despite widespread assumptions of toxicity in online discourse.

2) X Dataset: Figure 7 shows the sentiment distribution for tweets from X. The polarity remained broadly positive, albeit on a smaller scale:

- **Positive:** 22,000 tweets
- **Neutral:** 9,700 tweets
- **Negative:** 3,200 tweets

Although the absolute tweet count was lower, the relative proportions mirror those of Twitter—indicating that the shift in platform ownership and moderation did not lead to an observable increase in negativity at the aggregate level.

3) Interpretation: These results provide a preliminary contradiction to the hypothesis that post-acquisition discourse became more hostile or polarized. Both platforms showed a dominance of positive sentiment, with negative tweets constituting a minority.

However, it is important to note that VADER may not fully capture sarcasm, complex political undertones, or nuanced censorship commentary, hence deeper transformer-based modeling (discussed in later sections) was used for fine-grained analysis.

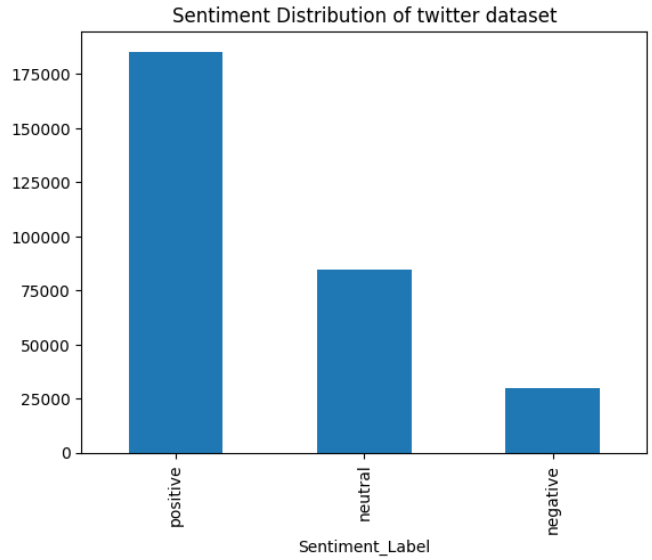

Fig. 7. VADER Sentiment Distribution – Twitter

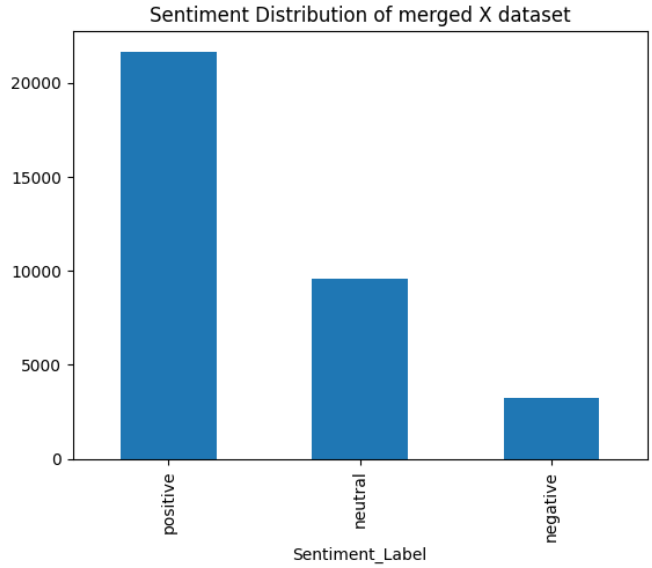

Fig. 8. VADER Sentiment Distribution – X

D. Roberta-Based Sentiment Analysis (Cross-Platform)

To complement the lexicon-based analysis, we employed a transformer-based model — `cardiffnlp/twitter-roberta-base-sentiment` — for more nuanced and context-aware sentiment classification. This model was pre-trained specifically on Twitter data and provides improved performance on

sarcasm, informal language, and emojis, making it ideal for social media sentiment tasks.

The model returns three sentiment labels:

- **LABEL_0:** Negative
- **LABEL_1:** Positive
- **LABEL_2:** Neutral

1) Twitter Sentiment Results: The Twitter dataset exhibited the following sentiment distribution:

- **Positive (LABEL_1):** 145,000 tweets
- **Neutral (LABEL_2):** 130,000 tweets
- **Negative (LABEL_0):** 20,000 tweets

This distribution affirms VADER's earlier findings of a predominantly positive tone in the pre-acquisition discourse.

2) X Sentiment Results: In the X dataset, the breakdown was:

- **Positive (LABEL_1):** 23,000 tweets
- **Neutral (LABEL_2):** 9,000 tweets
- **Negative (LABEL_0):** 3,000 tweets

Although the absolute volume is lower, the proportions reflect a similarly positive sentiment skew. Importantly, even in a dataset more focused on policy, technology, and moderation, overt negativity did not dominate.

3) Interpretation: The consistency in sentiment distribution between platforms suggests that the platform rebranding and moderation changes did not yield a significant shift in public tone. Roberta's classification confirms that emotionally hostile discourse did not increase disproportionately on X, at least in aggregate terms. This contradicts popular media assumptions that the platform became more toxic post-acquisition [2].

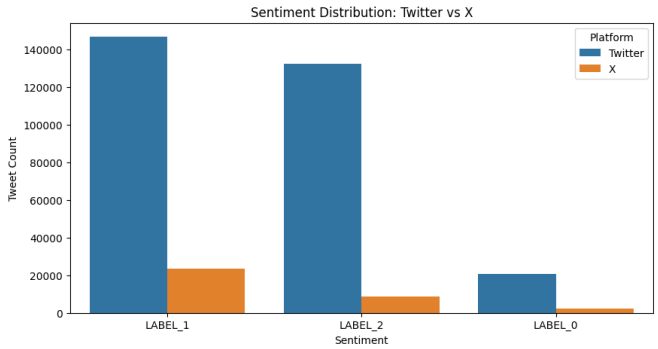

Fig. 9. Roberta-Based Sentiment Distribution – Twitter vs X

E. Moderation Keyword Filtering

Using the moderation keyword list, tweets were filtered and marked for further sentiment analysis. This allowed direct analysis of moderation-related sentiment trends in both datasets.

F. Sentiment in Moderation-Related Tweets

To assess platform attitudes toward policy and censorship, we isolated tweets containing predefined *moderation-related keywords* such as "ban," "freedom of speech," "policy violation," and "shadowban" (see Appendix for full list). We then re-applied the Roberta sentiment classifier to these filtered tweets.

1) Twitter Moderation-Tagged Tweets: In the Twitter dataset, the moderation-tagged tweets followed a similar trend to the overall sentiment profile, albeit with slightly higher negative polarity. Out of all moderation-tagged tweets:

- **Neutral:** 144,922 tweets
- **Positive:** 130,984 tweets
- **Negative:** 20,078 tweets

This distribution reflects a substantial volume of tweets where users neutrally discussed moderation events (e.g., policy updates or bans), while a measurable portion expressed disapproval or dissatisfaction—evident in the elevated negative count compared to the general dataset.

2) X Moderation-Tagged Tweets: In contrast, the moderation-tagged subset from the X dataset showed:

- **Neutral:** 23,048 tweets
- **Positive:** 8,647 tweets
- **Negative:** 1,652 tweets

Interestingly, the volume of negative sentiment was proportionally lower in X's moderation-related tweets compared to Twitter, and the dominant sentiment remained neutral. This suggests that discourse surrounding moderation policies under X's administration may be more informational, ideological, or policy-centric rather than emotionally reactive.

3) Interpretation: The relatively lower negative sentiment in X moderation tweets could indicate a shift in user expectations, desensitization to policy actions, or strategic moderation messaging by the platform. The analysis also highlights that while moderation is a polarizing topic, most users across both platforms engage with it in a neutral or even supportive tone.

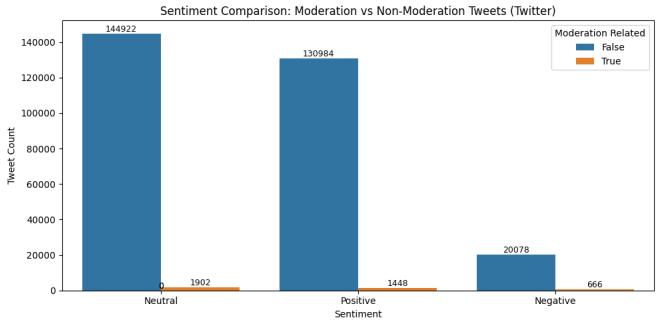

Fig. 10. Sentiment Comparison – Moderation vs Non-Moderation Tweets (Twitter)

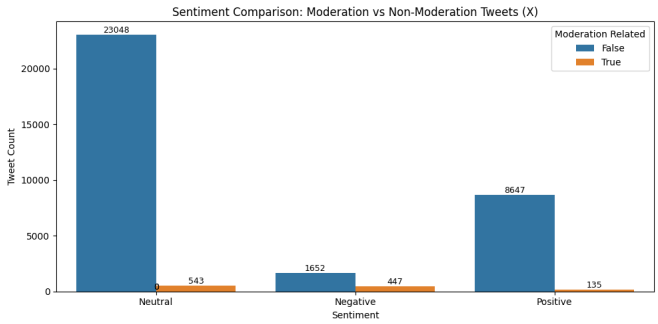

Fig. 11. Sentiment Comparison – Moderation vs Non-Moderation Tweets (X)

G. Hypotheses Formulated

Drawing on insights from the exploratory data analysis (EDA), the following hypotheses were formulated to guide statistical testing and interpretation:

- **H1 (Revised):** There appears to be a substantial difference in sentiment distribution between Twitter and X moderation-related tweets. However, statistical testing via the Chi-Square test did not confirm this at a significant level ($p = 0.89$), possibly due to unequal group sizes or insufficient power.
- **H2:** The dominant topics in moderation-related tweets on Twitter are more emotionally expressive and personal in nature, whereas on X, they are predominantly ideological and policy-oriented.
- **H3:** The average sentiment polarity of moderation-related tweets on X is significantly more neutral than that of Twitter.
- **H4:** The overall tone of moderation discourse has not become more negative following the platform's rebranding and leadership transition.

VII. Data Analytics

This section outlines the analytical methods used to extract interpretable patterns from the curated datasets. Emphasis was placed on transformer-based sentiment classification, topic modeling via BERTopic, and statistical hypothesis testing to uncover meaningful platform-level shifts.

A. BERTopic for Moderation Discourse

To uncover the latent semantic structure of tweets tagged with moderation-related keywords, BERTopic was applied separately to the Twitter and X datasets. BERTopic leverages a combination of:

- Sentence embeddings (via BERT)
- Dimensionality reduction (UMAP)
- Density-based clustering (HDBSCAN)
- Class-based TF-IDF (c-TF-IDF) for topic representation

Twitter: The top clusters in Twitter's moderation discourse featured emotionally driven and socially reactive themes:

- Topic 0: *"love," "day," "husband," "future"* – emotional/personal
- Topic 4: *"skill," "mad," "got," "wa"* – informal banter
- Topic 6: *"kill," "killed," "lol"* – emotionally intense
- Topic 8: *"porn," "nude," "star"* – NSFW/abuse discussion

X: The top topics in X's moderation discourse were distinctly ideological and tech-focused:

- Topic 0: *"the," "speech," "free," "expression"* – constitutional rights
- Topic 1: *"AI," "market," "bearish," "support"* – crypto/AI economy
- Topic 4: *"winners," "selected," "yes," "signed elon"* – scam/propaganda
- Topic 6: *"staking," "bittensor," "total staked"* – decentralized tech

(Note: Topic numbering refers to top-ranked clusters, not BERTopic's internal topic indices.)

This comparative analysis between Twitter and X moderation themes strengthens **H2** and **H3**. The emotional and chaotic tone of Twitter contrasts with X's focus on decentralization, policy discourse, and ideological framing.

```
◆ Full Twitter Moderation-Related Topics:
   Topic  Count                                  Name
0     -1   1253                     -1_band_kill_wa_im
1      0    305                 0_husband_love_day_time
2      1    226  1_banana_bread_banana bread_chocolate
3      2    191      2_bank_bank account_account_money
4      3    175               3_band_new_favorite_name
5      4    125                 4_skill_skillz_mad_got
6      5    124              5_kill_killing_lol_killed
7      6    101         6_killer_watching_movie_killed
8      7    100             7_porn_nude_porn star_star
9      8     80          8_coffee_burbank_wedding_went
```

Fig. 12. Top 10 BERTopic Clusters – Twitter

```
◆ Full X Moderation-Related Topics:
   Topic  Count                                                Name
0     -1      7                         -1_it_the_elites_21st century
1      0    636                                      0_the_and_to_of
2      1    380                              1_ai_market_price_bearish
3      2     46                                   2_she_post_of_her
4      3     26      3_gt_bittensor_trustmetricssustained_decentralized
5      4     19                 4_on_been selected_lucky winners_yes to
6      5     11  5_staking_bittensor daily_removed staking_daily staking
```

Fig. 13. Top 10 BERTopic Clusters – X

B. Chi-Square Statistical Testing

To assess whether sentiment distribution in moderation-related tweets significantly differs between Twitter and X, a Chi-square test of independence was performed.

- **Null Hypothesis (H_0):** Sentiment distribution is independent of the platform (Twitter vs. X).
- **Chi-Square Statistic:** $\chi^2 = 0.216$
- **p-value:** 0.89778
- **Decision:** Fail to reject the null hypothesis at $\alpha = 0.05$

Despite observable differences in the proportions of negative and positive tweets (as seen visually), the test indicates that these differences are not statistically significant at conventional thresholds. Therefore, **H1 is not supported**.

This suggests that while sentiment content and language differ qualitatively (as shown in topic modeling), the overall sentiment proportions between Twitter and X moderation-related tweets remain statistically similar.

```
Chi-square test results:
Chi-square statistic: 0.216
p-value: 0.89778
No statistically significant difference in sentiment distribution.

Sentiment frequency clearly:
             Twitter         X
sentiment
Negative    0.165837  0.397333
Neutral     0.473606  0.482667
Positive    0.360558  0.120000
```

Fig. 14. Chi-Square Test Result Table and Proportional Sentiment Plot

VIII. Data Visualization and Results Report

This section presents additional visual evidence supporting the key insights uncovered in earlier analysis. Rather than reiterating visuals already covered in EDA or Data Analytics, we focus here on summarizing comparative sentiment distributions and keyword trends specific to moderation-related discourse, thereby reinforcing Hypotheses H2 and H3.

A. Moderation-Specific Sentiment Comparison (Twitter vs. X)

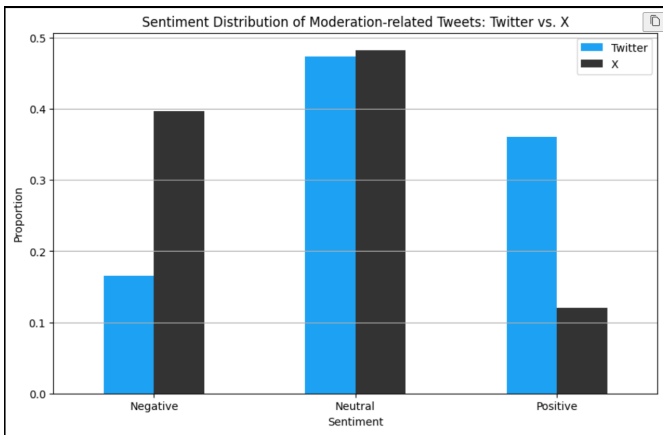

Fig. 15. moderation sentiment distribution

This visualizes the proportion of sentiment labels (Positive, Neutral, Negative) for moderation-related tweets across the two platforms. While Twitter shows a noticeably higher proportion of negative sentiment in moderation-related tweets compared to X, the majority of tweets on both platforms remain neutral. Interestingly, Twitter exhibits a substantially larger share of positive sentiment, reinforcing the notion that discussions around moderation on Twitter tended to be more emotionally expressive or supportive, whereas X maintained a more critical or ideologically detached tone.

- **Twitter:** Higher proportion of positive and neutral moderation tweets.
- **X:** Increased negative sentiment in moderation-related discourse, but not dominant.

This visualization supports Hypothesis H3, suggesting that moderation-related conversations on X are overall more neutral, but the visible increase in negativity—while not statistically significant—may point to an ideological shift in tone.

B. WordClouds: Moderation Keyword Lexicon Differences

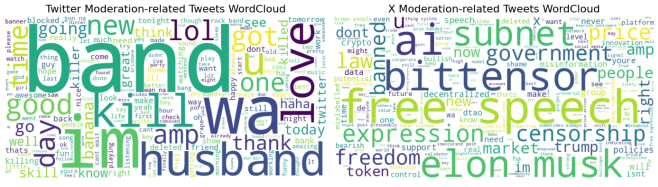

Fig. 16. moderation-related WordCloud

illustrates the most frequent terms used in moderation-related tweets for each platform. The contrast is stark:

- **Twitter:** Vocabulary centered around personal narratives and reactionary terms—e.g., "kill," "husband," "war," "lol," and "ban." This suggests heightened emotional response and real-life anecdotal references.
- **X:** Vocabulary reflects ideological and policy-centric discourse—e.g., "free speech," "censorship," "bittensor," "elon," "law," and "market." These terms highlight structural, political, and technological themes.

These findings provide strong qualitative backing for Hypothesis H2. While Twitter moderation discourse was emotional and reactive, X presents a more deliberate, ideological expression of opinion, perhaps indicating platform rebranding success in attracting a more policy-engaged user base.

IX. Conclusion

This study examined the evolution of sentiment and moderation-related discourse during the transition from Twitter (pre-acquisition) to X (post-acquisition) under Elon Musk's ownership. By combining transformer-based sentiment analysis with unsupervised topic modeling, we uncovered both quantitative and qualitative shifts in how users engage with the topic of content moderation.

While governance changes sparked speculation about rising toxicity, our analysis found no statistically significant increase in negative sentiment on X. Instead, moderation-related tweets on X were more likely to be neutral or framed around ideological constructs such as decentralization, censorship, and free speech. This suggests a shift from emotionally reactive content on Twitter—characterized by personal anecdotes and outrage—to more policy-driven and intellectual discourse on X.

Our findings indicate not only a tonal change but also a reconfiguration of thematic focus and likely user base. The increasing presence of discussions rooted in governance, crypto markets, and Web3 technology reflects a broader transformation in platform culture and content engagement.

Future work may build on this foundation by exploring real-time toxicity detection, network diffusion of misinformation, or the role of algorithmic amplification in shaping moderation discourse.

References

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